

Aldridge Alegre

+1 (925) 250-1305 github.com/alalegre linkedin.com/in/alalegre almy.alegre@gmail.com

Education

University of California, Santa Cruz

B.S. in Computer Science

Santa Cruz, CA

Sep 2022 – June 2026

Experience

Rocket Team | Ground Control Systems – Frontend Developer

Sep 2024 – Present

- Collaborated with multidisciplinary subteams to design and maintain internal software tools supporting rocketry projects
- Improved the club's website structure and functionality to enhance usability for technical and non-technical members
- Authored and organized technical content describing subteam responsibilities using Markdown
- Developed a React-based timestamp visualization page to display flight-data metrics, improving cross-team data analysis workflows
- Contributed clean, maintainable code while iterating on designs based on teammate feedback

Technical Projects

Fullstack Email Web App | *React, Material UI, Node.js, Express.js, Docker, PostgreSQL, OpenAPI*

Feb 2026

- Built a fullstack web app that dynamically rendered 100 emails per mailbox by using React and Material UI, backed by a PostgreSQL database containerized with Docker
- Utilized Node.js and Express.js to provide a RESTful API to retrieve, create, and move the information of 400 people across the database
- Utilized OpenAPI to validate and enforce strict request validation by defining a restrictive 3.0.3 schema, reducing the need for defensive JavaScript code
- Validated and tested all functionalities using Vitest, React Testing Library, and Supertest, achieving 100% code coverage

Email Dashboard | *React, Material UI, Vitest, React Testing Library*

Feb 2026

- Developed a responsive dashboard that dynamically showed 75 emails per mailbox by using React and Material UI, using a static JSON data source
- Utilized Material UI's built-in responsive components to deliver a seamless email reading experience across both mobile and desktop screen sizes
- Wrote unit and integration tests using Vitest and React Testing Library to validate component behavior and achieve 100% code coverage

Snowball Detector | *React, Recharts, Supabase*

Aug 2025 – Nov 2025

- Designed and implemented an interactive dashboard analyzing 30+ study-session entries in real time to identify under-prioritized courses
- Developed 10+ responsive components with React to present data trends and user insights
- Built data-driven visualizations using Recharts, demonstrating understanding of data structures and statistical patterns
- Integrated a Supabase backend to support persistent storage, scalable queries, and responsive UI updates

Learnivity | *React, Supabase*

Aug 2025

- Built a real-time community discussion platform supporting 30+ live updates through Supabase
- Implemented 3+ accessible, responsive React components focused on clean design and maintainability
- Integrated secure database operations and efficient client-server communication

Auto Recall CLI | *Python, Google Calendar API*

July 2025 – Aug 2025

- Developed a command-line spaced-repetition tool scheduling 10+ study events automatically through Google Calendar API
- Automated event creation, deletion, and filtering, cutting manual planning time by 80%+

Skills

Languages: Python, JavaScript, HTML/CSS, Markdown

Frameworks & Tools: React, Recharts, Material UI, Node.js, Express.js, OpenAPI, Docker, Vitest, Supertest, React Testing Library, Google Calendar API, Git/GitHub

Databases: PostgreSQL, Supabase